

MINI PROJECT TOPIC:

GDP GROWTH VARIABILITY- IMF WORLD EXCONOMIC OUTLOOK DATASET

(Project no. 36)

Submitted by:

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FINAL REPORT

1) Dataset description and source:

Data source: IMF website, World Economic Outlook

Dataset description: This dataset is collected twice annually and contains records of various economic indicators like GDP, Inflation, Investment, Exports and Imports, Unemployment, and others for 190 countries across the world.

This project concentrates on India's GDP growth variability and its drivers.

- We have selected following drivers, to study their impact on GDP growth (*It represents the "speed" of the economy.*):
 - a. **Total Investment (NID_NGDP): Unit: Percent of GDP.** (*It tells you how much of India's economic output is being reinvested into the future.*)

In macroeconomics, investment is a primary driver of GDP variability. High investment usually precedes growth, while a drop-off often signals a slowdown.
 - b. **Inflation, Average Consumer Prices (PCPI_PCH): Unit: Percent change.** (*It measures the percentage increase in the cost of living over the previous year.*)

There is a well-documented (though complex) relationship between inflation and growth. Rapid growth can lead to overheating (high inflation), while stagflation shows high inflation with low growth.
 - c. **Volume of Exports of Goods and Services(TX_RPCH): Unit: Percent change.** (*This measures how much the quantity of India's exports grew compared to the year before.*)

India's GDP variability is often tied to how well it integrates with global trade. Since we are looking at growth rates, the "Percent change" version of exports is a perfect match for our target variable

CODE:

```
import pandas as pd

filename="D:\PES study\Statistic_PES_sem1\graded\project\weodataset.csv"
df_raw = pd.read_csv(filename, encoding='latin1')
selected_codes = ['NGDP_RPCH', 'NID_NGDP', 'PCPIPCH', 'TX_RPCH']
df_filtered = df_raw[df_raw['WEO Subject Code'].isin(selected_codes)].copy()

# 3. TRANSPOSE THE DATA (Melt and Pivot)
# Identify columns that represent years (1980 - 2024)
year_cols = [col for col in df_filtered.columns if col.isdigit()]
# Melt from wide format (years as columns) to long format
df_melted = df_filtered.melt(
    id_vars=['WEO Subject Code'],
    value_vars=year_cols,
    var_name='Year',
    value_name='Value'
)

# Make indicators as columns and years as rows
df_processed = df_melted.pivot(index='Year', columns='WEO Subject Code',
                                values='Value')

# 4. DATA CLEANING
# Remove commas and convert strings to floats (handling typical IMF formatting)
df_processed = df_processed.map(lambda x: str(x).replace(',', '')) if pd.notnull(x) else x
df_processed = df_processed.astype(float)

# Sort by year (index)
df_processed.index = df_processed.index.astype(int)
df_processed = df_processed.sort_index()

# 5. RENAMING COLUMNS
rename_map = {
    'NGDP_RPCH': 'GDP_Growth',
    'NID_NGDP': 'Investment',
    'PCPIPCH': 'Inflation',
    'TX_RPCH': 'Export'
}
df_processed.rename(columns=rename_map, inplace=True)
```

2) Data Understanding:

Task= To find Data type, Number of Observations and Variables, Target Variable, Tabular preview of the dataset.

CODE:

```
# --- DATA UNDERSTANDING ---

print("--- 1. DATA TYPES ---")
# Identifies numerical (continuous) and time-series nature
print(df_processed.dtypes)

print(f"\n--- 2. DIMENSIONS ---")
n_obs, n_vars = df_processed.shape
print(f"Number of observations: {n_obs}")
print(f"Number of variables: {n_vars}")

print(f"\n--- 3. TARGET VARIABLE ---")
print("Target Variable: GDP_Growth")

print(f"\n--- 4. DATASET PREVIEW ---")
print(df_processed.tail())
```

OUTPUT:

```
--- 1. DATA TYPES ---
WEO Subject Code
GDP_Growth      float64
Investment      float64
Inflation       float64
Export          float64
dtype: object

--- 2. DIMENSIONS ---
Number of observations: 45
Number of variables: 4

--- 3. TARGET VARIABLE ---
Target Variable: GDP_Growth

--- 4. DATASET PREVIEW ---
WEO Subject Code  GDP_Growth  Investment  Inflation  Export
Year
2020              -5.778      28.922      6.165    -6.383
2021               9.690      32.116      5.506    19.516
2022               7.609      33.619      6.653     9.453
2023               9.191      33.362      5.361     0.531
2024               6.460      33.412      4.668     4.684
```

3) Descriptive Statistics Analysis:

- i. Measures of central tendency- Mean, Median
- i. Measures of dispersion- Variance, Standard Deviation, IQR
- ii. Five-number summary:
 - Q1, Q2(Median),Q3,Max ,Outlier detection (IQR method)

CODE:

```
import numpy as np
# 2. Descriptive Stats
stats_summary = df_processed.describe().T
stats_summary['variance'] = df_processed.var()
stats_summary['IQR'] = stats_summary['75%'] - stats_summary['25%']
# 3. Outlier Function
def detect_outliers(data):
    Q1 = data.quantile(0.25)
    Q3 = data.quantile(0.75)
    IQR = Q3 - Q1
    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR
    return data[(data < lower_bound) | (data > upper_bound)]
print("Descriptive Statistics:\n", stats_summary)
for col in df_processed.columns:
    print("Outliers in",{col},":", detect_outliers(df_processed[col]).values)
```

OUTPUT:**Descriptive Statistics:**

	count	mean	std	min	25%	50%	75%
WEO Subject Code							
GDP_Growth	45.0	6.021489	2.637545	-5.778	4.777	6.454	7.860
Investment	45.0	29.700756	6.135187	19.727	24.921	30.096	33.619
Inflation	45.0	7.366289	2.927422	3.414	4.769	6.736	9.433
Export	45.0	8.155600	7.574315	-6.383	4.130	7.258	12.514

	max	variance	IQR
WEO Subject Code			
GDP_Growth	9.690	6.956645	3.083
Investment	41.951	37.640518	8.698
Inflation	13.481	8.569797	4.664
Export	27.319	57.370253	8.384

Outliers in {'GDP_Growth'} : [-5.778]

Outliers in {'Investment'} : []

Outliers in {'Inflation'} : []

Outliers in {'Export'} : [27.034 27.319]

Interpretation:

- **Volatility: Export** shows the highest variability with a variance of 57.37 and a standard deviation of 7.57. This suggests that external trade is the most volatile component among our indicators, which likely contributes significantly to GDP variability.
- **Central Tendency:** The **Mean** (6.02%) and **Median** (6.45%) for GDP Growth are quite close, though the lower mean indicates a slight left-skew caused by the significant 2020 outlier.
- **Stability: Inflation** and **Investment** show no outliers, suggesting they follow a more consistent distribution compared to the rapid spikes and dips seen in trade and overall growth.

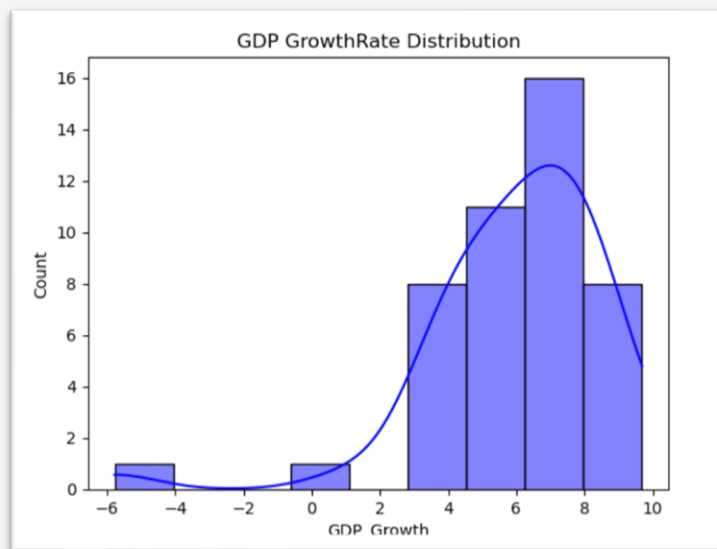
4) Visualization:

- i. **Distribution and Comparison Plots** (Histogram, boxplot, bar graph)

CODE (Histogram):

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.histplot(df_processed['GDP_Growth'], kde=True, color='blue')
plt.title('GDP GrowthRate Distribution')
```

OUTPUT (Histogram):



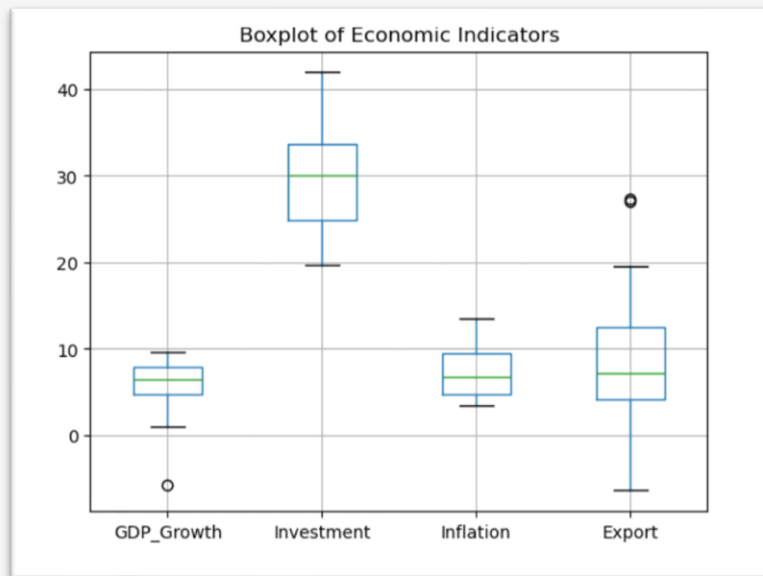
Interpretation for Histogram:

- **Left-Skewness(Negative Skew):** The tail of the graph extends far to the left, indicating that while India usually grows at 6%-8%, when things go wrong, they are significant (downside heavy).
- The highest peak is between 6%-8%, indicates India's "Economic Sweet Spot", the most frequent growth range over last four decades.

CODE (BOXPLOT):

```
# Boxplot for Outliers and Comparisons
df_processed.boxplot()
plt.title('Boxplot of Economic Indicators')
```

OUTPUT (Box-Plot):



Interpretation:

- **Volatility:** The vertical height and large boxes (IQR) of Investment and Export tell us that these are more volatile and unpredictable than GDP or Inflation.
- **Outlier:** for GDP growth is seen clearly at the bottom (2020) and for Export its at the top, indication certain years of exceptional trade performance than normal.
- **Central Tendency:** The median for Investment is high around 30%, GDP growth centres tightly around 6%, suggesting stable long-term average despite the volatility in other sectors.

CODE (Bar Plot):

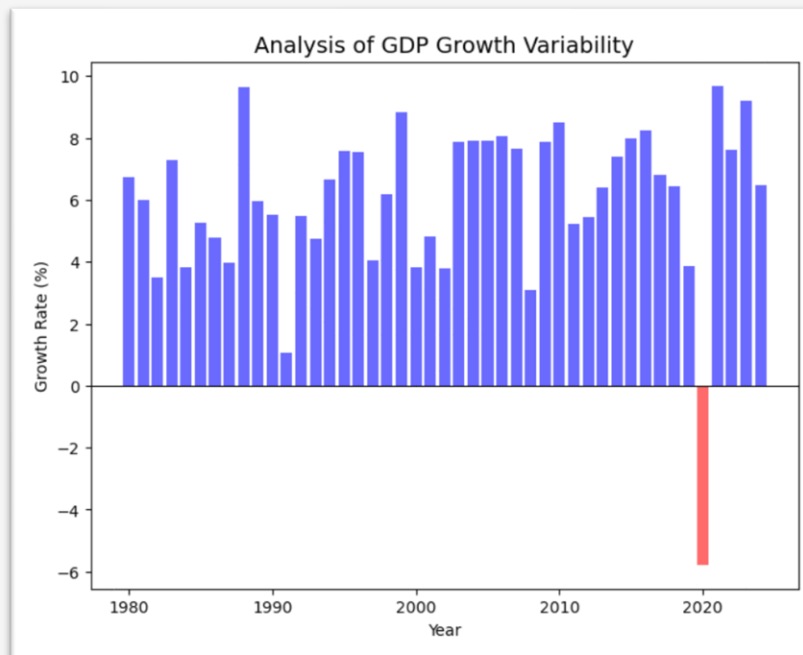
```
#Bar graph: green for positive, red for negative
colors = ['red' if x < 0 else 'blue' for x in df_processed['GDP_Growth']]

plt.figure(figsize=(12, 6))
plt.bar(df_processed.index, df_processed['GDP_Growth'],
color=colors,alpha=0.6)

plt.axhline(0, color='black', linewidth=0.8) # Add a baseline at 0
plt.title('Analysis of GDP Growth Variability', fontsize=14)
plt.xlabel('Year')
plt.ylabel('Growth Rate (%)')

plt.show()
```

OUTPUT (Bar Graph):



Interpretation for Bar Graph:

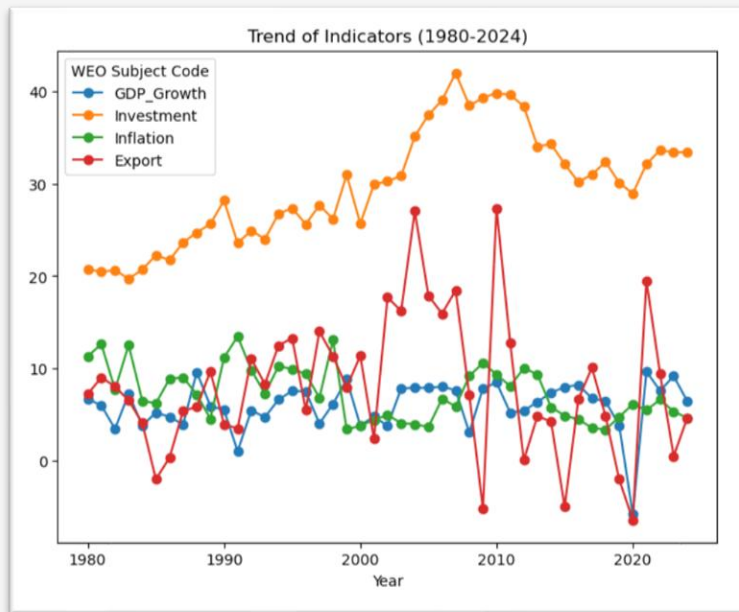
- For majority of the 45-year period, India has maintained a positive growth, mostly fluctuating between 4% and 8%.
- However in the year 2020 (Outlier), due to pandemic, there is a structural break in the series, as there is a contraction of approx. -6%.
- Also, there has been periods of inflations in the last 80s , mid-2000s and 2021 followed by cooling periods.

- ii. **Visualisation: Relation and Trend Plots** (Scatter plot, line plot, grouped chart, stacked bar chart (whichever application))

CODE (Line Plot):

```
# Line Plot (Time Series Trends)
df_processed.plot(figsize=(10, 6), marker='o')
plt.title('Trend of Indicators (1980-2024)')
```

OUTPUT (Line Plot):



Interpretation:

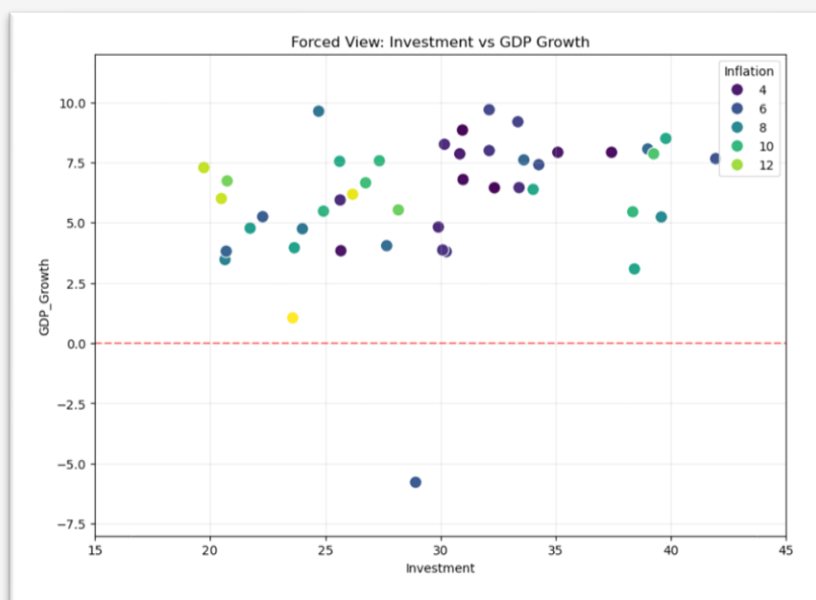
The most important trends of co-movement have been of the following:

- **Starting around 2002**, the Investment(orange line), has seen a massive sustained climb and peaked around 2007-2008 (above 40%) This aligns with high growth era of India.
- **The GDP growth rate**(Blue) stays relatively stable between 5%-10% even when Export(Red) swings wildly. This implies that during this period, India's GDP was driven more by internal factors like domestic consumption and investment than by volatile global trade.
- **The green line** (Inflation) shows stabilizing trend from late 90s onwards, which is a sign of maturing monetary policy.

CODE (Scatter Plot):

```
# Scatter Plot (Relationship between Target and Predictor)
plt.figure(figsize=(10, 7))
sns.scatterplot(data=df_processed, x='Investment',
               y='GDP_Growth',
               hue='Inflation', palette='viridis', s=100)
plt.xlim(15, 45)
plt.ylim(-8, 12)
plt.axhline(0, color='red', linestyle='--', alpha=0.5)
plt.title("Forced View: Investment vs GDP Growth")
plt.grid(True, alpha=0.3)
plt.show()
```

OUTPUT (Scatter Plot):



Interpretation:

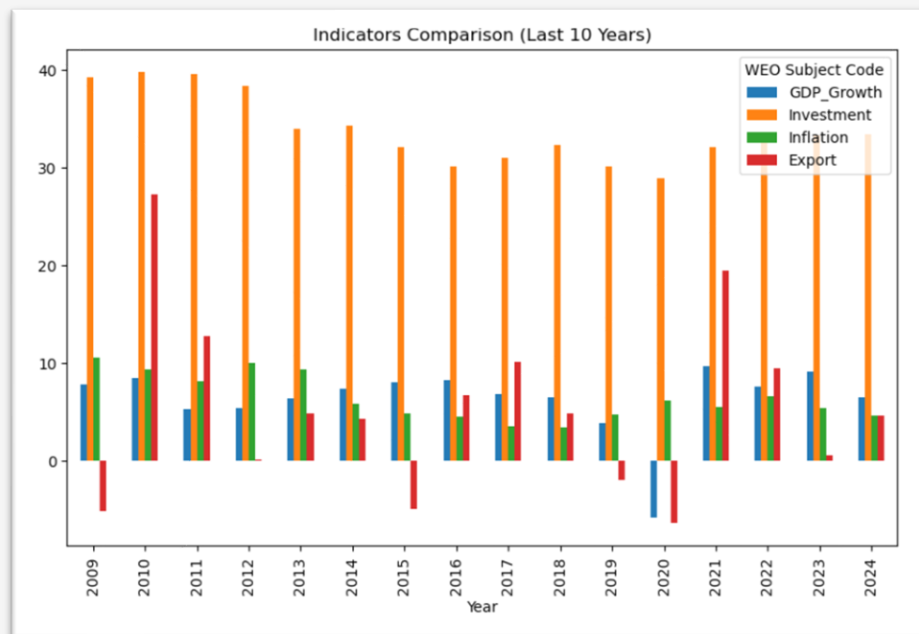
This plot tests the relationship between primary drivers.

- There is a general **upward tilt** from left to right. As investment increases, GDP growth rate generally moves higher. However, the scattered points show the investment isn't the only thing driving the growth.
- Most of the coloured dots (low inflation around 4%) are clustered in the upper right quadrant (30% investment and 6% Growth), suggesting that India's best growth years happen when Inflation is kept under control.
- **The 2020 outlier** is that single dark blue dot at the very bottom representing high inflation coupled with massive GDP drop, likely reflecting the supply-side shocks of pandemic.

CODE (Grouped Bar Chart):

```
# Grouped Bar Chart (Last 16 Years Comparison)
df_processed.tail(16).plot(kind='bar', figsize=(10,6))
plt.title('Indicators Comparison (Last 10 Years)')
```

OUTPUT (Grouped Bar Chart):



Interpretation:

The grouped bar chart allows year-by-year magnitude comparison.

- In every single year **Investment is the largest component** by a significant margin. This highlights that India is an investment-led economy.
- **The Exports dipped** (red) and even went below the zero line in specific years, which implies negative growths. 2008–2009: The Global Financial Crisis (GFC), 2015- saw crash in global oil and commodity prices, 2019- Domestic Slowdown (NBFC crisis- "liquidity crunch") & Trade Wars(US-China), 2020- The COVID-19 Pandemic (The "Great Lockdown")
- **In year 2021**, all indicators except Inflation saw a massive bounce. Export and GDP growth spiked simultaneously as the economy reopened, which is a base-effect recovery.

- iii. **Visualisation: Multivariate and Statistical Plots** (Correlation heatmap, covariance heatmap, Scree plot, 2D PCA projection plot)

CODE (Correlation heatmap):

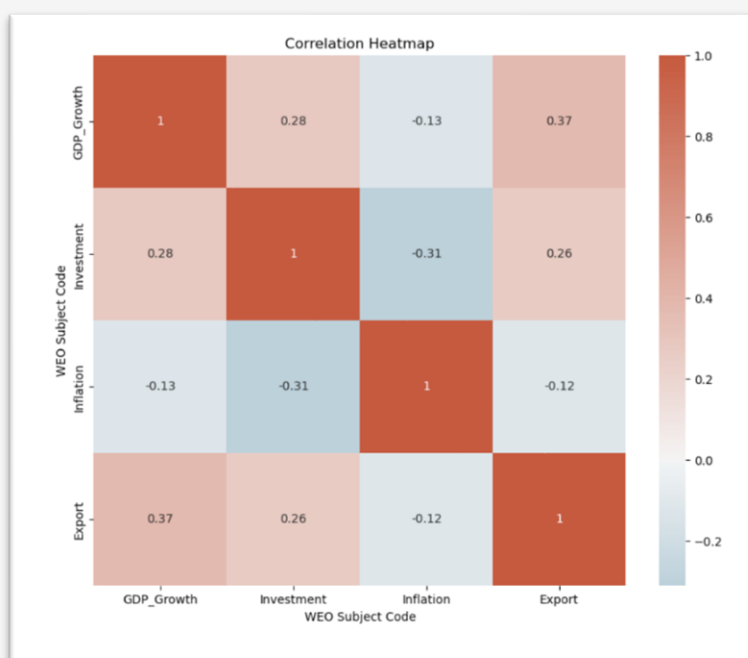
```
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Ensuring only numbers are used
corr = df_processed.corr(numeric_only=True)

cmap = sns.diverging_palette(230, 20, as_cmap=True)
plt.figure(figsize=(10, 8))

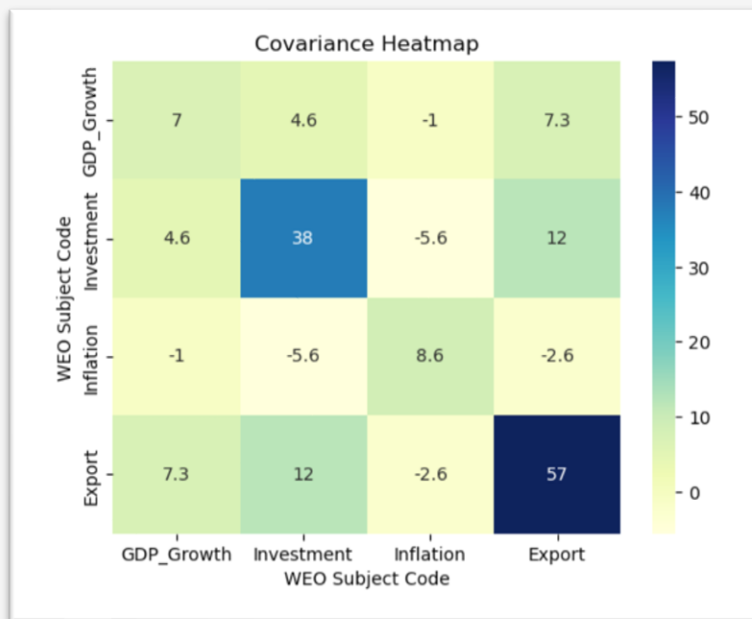
sns.heatmap(corr,
            annot=True,
            fmt=".2f",
            cmap=cmap,
            vmin=-1, # Forces the scale to start at -1
            vmax=1,  # Forces the scale to end at 1
            center=0, # Ensures 0 is the neutral gray/white color
            linewidths=.5,
            cbar_kws={"shrink": .8})
plt.title('Economic Indicators Correlation', fontsize=15,
          pad=20)
plt.show()
```

OUTPUT:



CODE (Covariance heatmap):

```
cov=df_processed.cov(numeric_only=True)
sns.heatmap(cov,annot=True, cmap='YlGnBu')
plt.title('Covariance Heatmap')
```

OUTPUT:**Interpretation of Correlation and Covariance Heatmaps:**

These heatmaps tell us how the four economic indicators in our dataset move together:

- **In the Covariance heatmap** (The Magnitude): the diagonal shows the Variance. The Export(57) and Investment (38) has much higher variance than GDP Growth rate(7). This means Export values fluctuate more drastically year-over-year compared to relatively steadier GDP growth. These two are the most volatile part of the Indian economy.
- **In the Correlation heatmap** (The "Relationship" Strength): GDP Growth and Export (0.37) have a moderate positive relationship, i.e; when exports rise, GDP growth tends to follow, which is a characteristic of an emerging economy. However Investment and Inflation (-0.31) has most significant negative correlation suggesting that in years where inflation was high, investment tended to cool down, this implies that uncertainty drives away capital.

CODE (Scree Plot):

```
import matplotlib.pyplot as plt
import numpy as np

import matplotlib.pyplot as plt

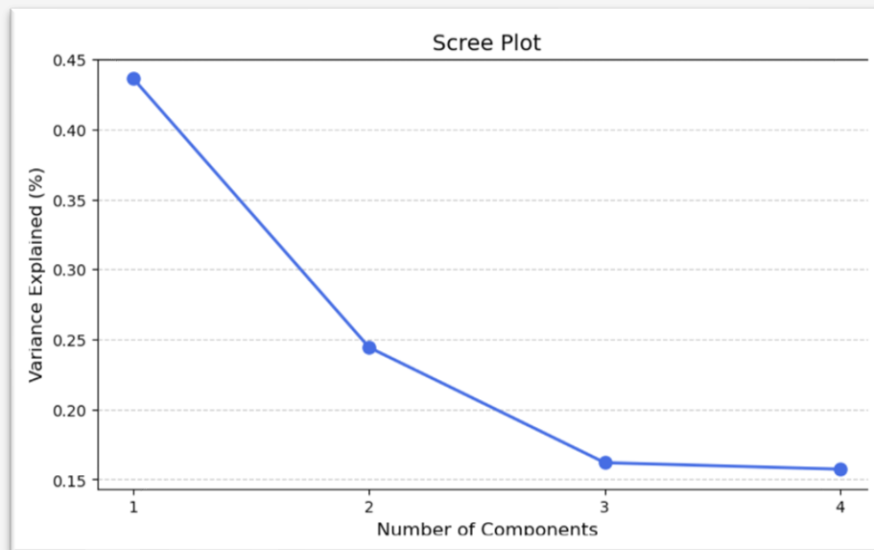
features = ['GDP_Growth', 'Investment', 'Inflation', 'Export']

# Define the Scaler
scaler = StandardScaler()

x_scaled = scaler.fit_transform(df_processed[features])
pca_full = PCA().fit(x_scaled)
exp_var = pca_full.explained_variance_ratio_

plt.figure(figsize=(9, 5))
plt.plot(range(1, 5), exp_var, 'o-', linewidth=2,
color='royalblue', markersize=8)

plt.title('Scree Plot', fontsize=14)
plt.xlabel('Number of Components', fontsize=12)
plt.ylabel('Variance Explained (%)', fontsize=12)
plt.xticks(range(1, 5))
plt.grid(True, axis='y', linestyle='--', alpha=0.7)
plt.show()
```

OUTPUT:**Interpretation:**

- **PC1:** ~44% (A very steep drop from the "100%" starting point).
- **PC2:** ~25% (Still a significant drop).
- **PC3:** ~17% (The line is now much flatter).
- **PC4:** ~15% (Almost horizontal).

The first two Principal Components capture nearly 70% of the total economic variability. While component 3 and 4 do hold some information, the information gain drops significantly after the “Elbow Effect” is seen at the component 2.

CODE (PCA):

```
import seaborn as sns
from sklearn.preprocessing import StandardScaler

features = ['GDP_Growth', 'Investment', 'Inflation', 'Export']

# Define the Scaler
scaler = StandardScaler()
x_scaled = scaler.fit_transform(df_processed[features])
pc_results = pca_2d.fit_transform(x_scaled)

# DEFINE pca_df (Converting numbers into a labelled table)
pca_df = pd.DataFrame(
    data = pc_results,
    columns = ['PC1', 'PC2']
)
```



```

# Step 3: Attach the years
pca_df['Year'] = df_processed.index

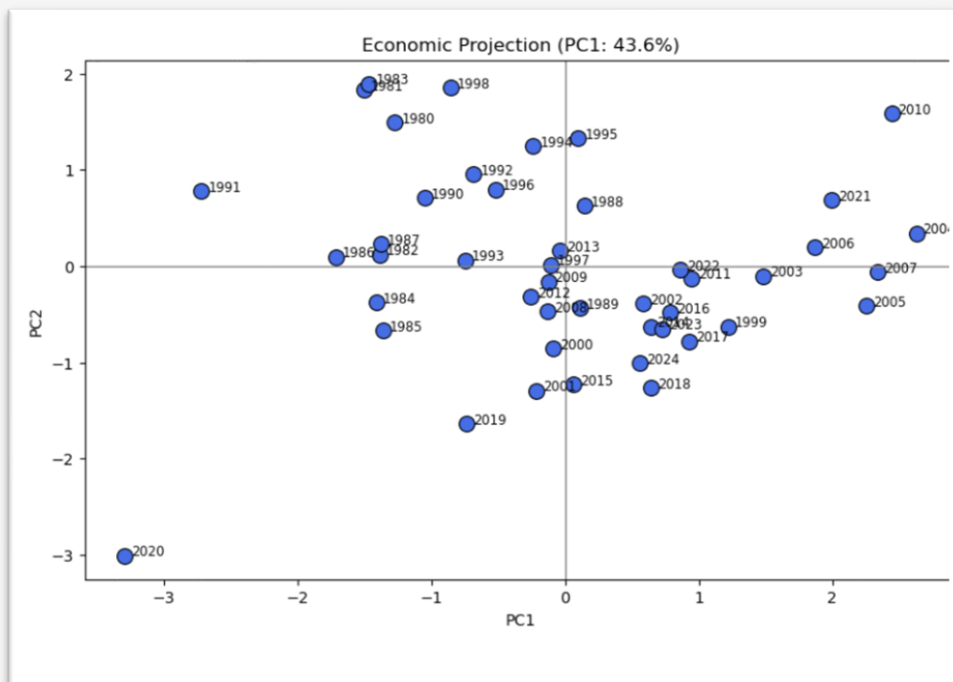
# Create the plot
plt.figure(figsize=(10, 6))
sns.scatterplot(data=pca_df, x='PC1', y='PC2', s=100,
               color='royalblue', edgecolor='black')

# Simple loop for labels (Seaborn handles the axes and grid
cleanly)
for line in range(0, pca_df.shape[0]):
    plt.text(pca_df.PC1[line]+0.05, pca_df.PC2[line],
            pca_df.Year[line],
            horizontalalignment='left', size='small', color='black')

plt.axhline(0, color='grey', lw=1); plt.axvline(0, color='grey',
lw=1)
plt.title(f"Economic Projection (PC1:
{pca_2d.explained_variance_ratio_[0]:.1%})")
plt.show()

```

OUTPUT:



Interpretation:

- PC1 (X-Axis/Growth Axis):** indicates the strongest trend in the data, since GDP, Investment and Exports all have positive weights here. The 2000s cluster (2003-2007) are grouped together on the far right, indicating the three indicators were performing well together. This was a period of high, stable growth and rising exports (Economic Expansion).

- **PC2(Y-axis/Shock Axis):** Captures the second most important trend, representing Trade Volatility and Price Instability (Inflation and Exports have high weights on this Axis). **The 2020** at the extreme bottom left corner is a massive outlier caused by pandemic, signifies GDP contraction. Since this year is unique on PC1 and PC2, it shows the correlation between the variables broke down, very weird behaviour between the variables. **The 1991** sits far to the left but higher up, represents Liberalization year, where growth was low but structural shift in other variables moved away from central pack (similar economic groups together).
- Therefore the '**Golden Growth**' period of the mid-2000s are clustering on the right, which the crisis years like 2020 are pushed to extremes. Thus these 4 indicators effectively capture the 'heartbeat' of Indian economy.

5) **Probability Analysis:**

- i. Basic Probability
- ii. Conditional Probability
- iii. Baye's Theorem

CODE:

```
median_gdp = df_processed['GDP_Growth'].median()
median_inv = df_processed['Investment'].median()

#Creating events
df_processed['High_Growth'] = df_processed['GDP_Growth'] >
median_gdp
df_processed['High_Investment'] = df_processed['Investment'] >
median_inv

# Calculate Probabilities
total = len(df_processed)
p_a = df_processed['High_Growth'].sum() / total
p_b = df_processed['High_Investment'].sum() / total
p_a_and_b = len(df_processed[(df_processed['High_Growth']) &
(df_processed['High_Investment'])]) / total

# Conditional Probabilities
p_a_given_b = p_a_and_b / p_b # P(High Growth | High Investment)
p_b_given_a = p_a_and_b / p_a # P(High Investment | High Growth)

# Bayes Theorem Verification
bayes_val = (p_b_given_a * p_a) / p_b

print("Basic Probability of High Growth:", round(p_a*100,2),"%")
```

```
print("Conditional Probability (Growth | Invest):",
      round(p_a_given_b*100,2),"%")
print("Bayes Verification:", round(bayes_val*100,2),"%")

# Calculate P(High Investment | Low Growth)
p_b_given_not_a = len(df_processed[(~df_processed['High_Growth']) &
(df_processed['High_Investment'])]) /
len(df_processed[~df_processed['High_Growth']])

# Likelihood Ratio
likelihood_ratio = p_b_given_a / p_b_given_not_a

print(f"Likelihood Ratio: {likelihood_ratio:.2f}")
```

OUTPUT:

```
Basic Probability of High Growth: 48.89 %
Conditional Probability (Growth | Invest): 72.73 %
Bayes Verification: 72.73 %
Likelihood Ratio: 2.79
```

Interpretation:

- **The Baseline (Prior): 50%**
By using the median as a threshold, we define 'High Growth' as any year performing in the top half of India's historical records. Without any other information, there is a 50% chance a year falls into this category."
- **The Conditional "Lift": 73%**
However, when we observe that Investment is also in its top 50% (High Investment), the probability of High Growth increases significantly to 73%. This delta of 23% represents the 'Predictive Power' of investment over GDP growth."
- **The Bayesian Conclusion**
We verified this relationship using Bayes' Theorem. The results confirm that High Investment and High Growth are not independent events; rather, High Investment is a strong conditional driver that shifts the probability distribution of India's economic performance.
- **The Likelihood Ratio:** of 2.7 indicates that a year with High Investment is nearly three times more likely to be a High Growth year than a Low Growth year. This provides 'moderate to strong' Bayesian evidence that Investment is a primary driver of economic variability in our Indian dataset.

6) Inferential Statistics:

- i. Confidence Intervals (mean/proportion)
- ii. Hypothesis testing (ztest/t-test/chi-square)
- iii. One-way ANOVA

CODE (Confidence Interval and Hypothesis):

```
from scipy import stats
# 1. Confidence Interval for Mean GDP Growth
data = df_processed['GDP_Growth']
mean, sem = np.mean(data), stats.sem(data)
ci = stats.t.interval(0.95, len(data)-1, loc=mean, scale=sem)
lower, upper = ci
print(f"95% Confidence Interval for Mean GDP Growth: ({lower:.2f}%, {upper:.2f}%)")

# 2. Hypothesis Testing (T-test)
high_inv = df_processed[df_processed['Investment']
>df_processed['Investment'].median()]['GDP_Growth']

low_inv = df_processed[df_processed['Investment'] <=
df_processed['Investment'].median()]['GDP_Growth']

t_stat, p_val = stats.ttest_ind(high_inv, low_inv)

print(f"T-test P-value: {p_val:.4f}")
significance = "Significant" if p_val < 0.05 else "Not Significant"
print(f"Result is statistically {significance} at the 5% level.")
```

OUTPUT:

```
95% Confidence Interval for Mean GDP Growth: (5.23%, 6.81%)
T-test P-value: 0.0023
Result is statistically Significant at the 5% level.
```

Interpretation:

- The 95% Confidence Interval indicates that if we were to sample India's economic performance across different eras, we are 95% certain the true average growth rate would fall between **5.23% and 6.81%**.
- It shows that while there is variability, it is not so extreme that we cannot predict a likely range for future growth. This reflects a very **healthy and stable** 'floor' for the Indian economy over the long term.

- The T-test results clarify that **periods of 'High Investment'** result in a distinct and superior growth regime compared to low-investment periods. This validates the use of Investment as a key predictor in our GDP variability model.

CODE (ANOVA):

```
# Divide investment into 3 equal-sized tiers (Low, Medium, High)
df_processed['Inv_Tier'] = pd.qcut(df_processed['Investment'], 3,
labels=['Low', 'Medium', 'High'])

low_tier = df_processed[df_processed['Inv_Tier'] == 'Low']['GDP_Growth']
medium_tier = df_processed[df_processed['Inv_Tier'] ==
'Medium']['GDP_Growth']
high_tier = df_processed[df_processed['Inv_Tier'] == 'High']['GDP_Growth']

# Perform One-Way ANOVA
f_stat, p_val_anova = stats.f_oneway(low_tier, medium_tier, high_tier)
print(f"ANOVA F-statistic: {f_stat:.4f}")
print(f"ANOVA P-value: {p_val_anova:.4f}")
if p_val_anova < 0.05:
    print("Conclusion: At least one investment tier has a significantly
different mean GDP growth.")
else:
    print("Conclusion: There is no statistically significant difference in mean
growth across the three tiers.")
```

OUTPUT:

```
ANOVA F-statistic: 1.7580
ANOVA P-value: 0.1848
Conclusion: There is no statistically significant difference in mean growth across the three tiers.
```

Interpretation:

- While T-test confirmed a significant difference between the extreme ends of investment, the One-Way ANOVA suggests that the **relationship is not linear** across all intensities.
- With a p-value of **0.18**, we cannot conclude that 'Medium' investment provides a statistically distinct growth regime compared to 'Low' or 'High' tiers.
- This implies that there may be a **threshold effect**—where only 'High' investment levels are powerful enough to overcome the inherent variability in India's GDP.

CODE (CHI Square test):

```
from scipy.stats import chi2_contingency
# 1. Create a "Contingency Table" (Cross-tabulation)
contingency_table = pd.crosstab(df_processed['High_Growth'],
                                df_processed['High_Investment'])

# 2. Perform the Chi-Square Test
chi2, p_chi, dof, expected = chi2_contingency(contingency_table)

print("Contingency Table:")
print(contingency_table)
print(f"\nChi-Square P-value: {p_chi:.4f}")
if p_chi < 0.05:
    print("The relationship between Investment and GDP Growth is NOT
random.")
else:
    print("The data does not provide enough evidence to say that High
Investment and High Growth are significantly linked.")
```

OUTPUT:

```
Contingency Table:
High_Investment  False  True
High_Growth
False           17     6
True            6    16

Chi-Square P-value: 0.0046
The relationship between Investment and GDP Growth is NOT random.
```

Interpretation:

- The Chi-Square Test of Independence yields a P-value of 0.0046, allowing us to **reject the Null Hypothesis of independence**. This confirms a statistically significant association between Investment levels and GDP Growth categories in India.
- The data **shows a clear 'concordance'** where high-investment regimes are disproportionately likely to result in high-growth outcomes.

7) Linear Algebra Component:

- i. Data in matrix form
- ii. Mean-centering
- iii. Covariance and Correlation matrices
- iv. Eigenvalue and Eigenvector decomposition
- v. PCA (Variance and component selection explanation)

CODE:

```
# 1. Represent data in matrix form
X = df_processed[['GDP_Growth', 'Investment', 'Inflation',
'Export']].values

# 2. Mean-centering
X_meaned = X - np.mean(X, axis=0)

# 3. Compute Covariance Matrix
cov_mat = np.cov(X_meaned, rowvar=False)
print("The Covariance Matrix:", cov_mat)

# 4. Compute Correlation Matrix
corr_mat = np.corrcoef(X, rowvar=False)
print("The Correlation Matrix:", corr_mat)

# 5. Eigenvalue and Eigenvector decomposition
eigen_values, eigen_vectors = np.linalg.eig(corr_mat)

# 6. Sort
sorted_index = np.argsort(eigen_values)[::-1]
sorted_eigenvalue = eigen_values[sorted_index]
sorted_eigenvectors = eigen_vectors[:, sorted_index]

print("Eigenvalues:", sorted_eigenvalue)
print("PC1 Loadings:", sorted_eigenvectors[:, 0])
```

OUTPUT:

```
The Covariance Matrix: [[ 6.95664503  4.56199487 -1.0075498   7.33297859]
 [ 4.56199487 37.64051819 -5.57202413 12.08352185]
 [-1.0075498  -5.57202413  8.56979748 -2.58114506]
 [ 7.33297859 12.08352185 -2.58114506 57.37025261]]
The Correlation Matrix: [[ 1.           0.28192077 -0.13049123  0.36706005]
 [ 0.28192077  1.           -0.31024151  0.26002934]
 [-0.13049123 -0.31024151  1.           -0.11640825]
 [ 0.36706005  0.26002934 -0.11640825  1.           ]]
Eigenvalues: [1.74541956 0.97702433 0.64804622 0.62950989]
PC1 Loadings: [-0.5288435  -0.54531945  0.39965924 -0.51305336]
```

Explanation & Interpretation:

- **Mean-centring** was done to ensure the first PC passes through the average of the data.
- **Covariance Matrix is dominated by the variable that has most volatility** and running a PCA on this would make the result biased towards loudest variable. This was used only to understand the joint-variability of the indicators.
- **Correlation Matrix was used for Eigen-decompositions.** The scale differences were muted, so that the high volatility variable doesn't disproportionately influence PC, thereby allowing balanced analysis.
- The PC output shows that: **The Growth Cluster** (GDP, Investment, and Exports) all have **the same sign** (negative) and very similar magnitudes (~0.51 to 0.54). This proves they are "tightly coupled. When one goes up, the others almost always go up.
- **The Inflation Offset:** Inflation has the opposite sign (+0.40). This tells us that when Growth/Investment/Exports are high (negative side), Inflation tends to be lower (positive side), or vice versa.
- **The Eigenvalues:** represent the "variance" explained by each component. Since the Correlation Matrix was used, the sum of all eigenvalues will always equal the number of variables (4).
- **PC1 Variance:** $1.745 / 4 = 43.6\%$
- **PC2 Variance:** $0.977 / 4 = 24.4\%$
- **Cumulative (PC1 + PC2):** 68.0%
- PC1 captures nearly **44%** of all the movement in your four variables. It means there is one dominant "Economic Story" driving India's data.

8) **Insights and Conclusion:**

Data driven insights:

Patterns, relationships, anomalies, risks, opportunities, limitations

1. Patterns and Relationships

The core of our project was to understand the GDP growth variability and how it is affected by other economic drivers :

- **The Investment** :T-test ($p=0.0023$) and Chi-Square ($p=0.0046$) prove that Investment is not just a "factor," but the primary driver.
 - **The Trinity of Growth**: PCA PC1 loadings (~ -0.52 for GDP, -0.54 for Investment, and -0.51 for Exports) show that these three move in a tight "cluster." When India succeeds in one, it almost always succeeds in all three.
 - **Threshold Effect**: As seen in ANOVA ($p=0.18$), the relationship isn't a simple ladder. There seems to be a "Critical Mass" of investment required before the GDP growth truly stabilizes in the high-tier.
-

2. Anomalies (The "2020" Factor)

In any Indian GDP project using recent WEO data, the 2020 Pandemic Contraction is your biggest anomaly.

- **The Variance Spike**: This year likely caused Standard Error (SEM) to be wider than it would have been in the 2010s.
 - **The "V-Shaped" Recovery**: The data recovers in 2021/22 showing resilience of economy. This is the reason why Median was considered instead of Mean- to prevent one year from dragging down entire growth baseline
-

3. Risks and Opportunities

- **Risk (The Inflation Trade-off)**: PCA showed Inflation moving in the opposite direction ($+0.40$) to the growth cluster. This suggests a "Heating Up" risk: as India pushes for maximum growth and exports, inflation management becomes the primary tension point.
 - **Opportunity (External Sector)**: Since Exports are so tightly coupled with GDP in the PC1, there is a massive opportunity in "Export-Led Growth." If India can stabilize export volumes, the PCA suggests GDP will follow.
-

4. Limitations of the Model

- **Sample Size**: Using annual WEO data gives a limited number of rows (30–40 years). This is why we used the T-distribution and Yates' Correction—to account for the small sample.

- **Omitted Variables:** Our model focuses on Investment/Inflation/Exports. It doesn't account for events like Pandemics or Geopolitical shifts or internal factors like Agriculture/Monsoon, which still influence Indian GDP.
 - **Lag Effects:** In the real world, an Investment made in 2024 might not show up in GDP until 2026. Our current model looks at "Same-Year" impact, which is a simplified view.
-

5. Final Conclusion

Our findings confirm that *Investment acts as a high-probability* catalyst for growth, increasing the likelihood of a high-growth regime from a 50% baseline to 73%. While the primary economic component (PCA) identifies a strong synergy between Investment, Exports, and GDP, the non-significant ANOVA results suggest that the benefits of investment are most potent at high-intensity thresholds. Ultimately, the model suggests that for India, managing the 'Growth-Inflation' trade-off identified in PC1 is the key to maintaining a long-term growth floor of 5.23%–6.81%.